

TrES-3b

R-2566

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110406 · created 2026-08-26T23:34:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455657.98401 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.79 +/- 2.23 [%]
Orbital Inclination (inc)	81.48 +/- 2.86
Airmass coefficient 1 (a1)	1.92 +/- 0.41
Airmass coefficient 2 (a2)	-0.65 +/- 1.01
Scatter in the residuals of the lightcurve fit is	20.54 %
Best Comparison Star	#1 - [336, 340]
Optimal Aperture	2.03
Optimal Annulus	4.99
Transit Duration (day)	0.037 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.195 $\pm$ 0.057 vs 0.16309 (deviation 0.42 σ)
Duration fit vs published	0.037 d vs 0.0567 d (deviation 0.68 σ)
Mid-time O-C	0.7 min
Depth SNR	2.6
Residual scatter	20.54 %

Light curve

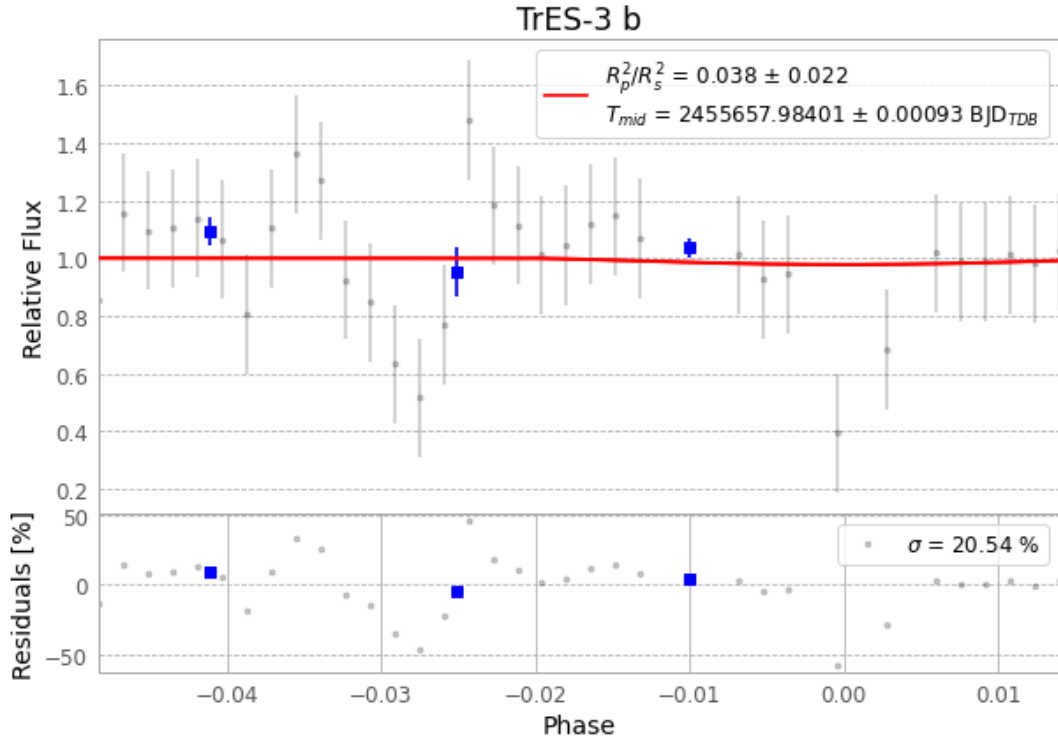


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-06.png (SHA-256 e58321b892530dc7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [233, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8148f9c8016c9109...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

40 FITS frames (26 MB), TRES-3110406100313.FITS → TRES-3110406120012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110406.log (a39d1ff65b70...), FinalLightCurve_TrES-3 b_2011-04-06.png (e58321b89253...), FinalLightCurve_TrES-3 b_2011-04-06.csv (08b283d934fa...)

Compute resources

Wall time 226.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:34Z.